

CAROLINA COMMUTE



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ITEC 444: Intro to Human Computer Int

Spring 2026

INITIAL IDEA



TRANSLOC – CURRENT SOLUTION

- Difficult to use interface
- Low update rate
- Very much a black box; no idea what is happening with buses
- Unstable; buses don't show up, don't update, etc



CAROLINA COMMUTE

- USC-specific campus transit app
- Modern Interface
- Bus info, status, notifications
- Route planning

GOALS



Give students more transparency

Inform students of “status” of bus

- Delayed
- Out of Service
- On Break



Higher refresh rate and indication

TransLoc is very slow
Give indication on when updates occur



Notifications

Give students notification of status changes
Notification of approaching stop

INTERFACE



Simplistic design

Easy to understand and low clutter



High-contrast

Easier for people with visual disabilities such as color-blindness



Icon+text

Quicker recognition of action



Large text

Easier for people with visual disabilities to read

USER POPULATION

- Intended Audience -
 - USC students
 - This app is only for the USC routes, which are only accessible by students
 - Regardless of if they use bus or walk, route planning is useful for timing
 - Off-campus students
 - Some parking lots are far from campus
 - Require bus to get from parking to campus



TASK ANALYSIS

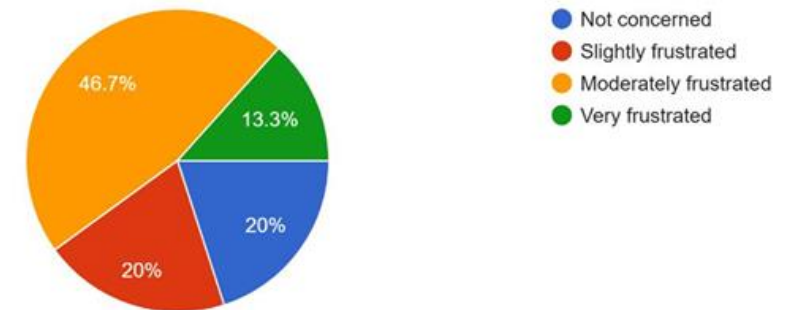
- How users use the current solution
 - How often do they use the system
 - How do they complete the tasks required to use the system
- What users don't like
 - Are there any issues with the current system?
 - What are they?
- What do they want to see?
 - What features or changes would users like to see?
 - How could we make it easier?
- These insights were used to identify user pain points and guide our survey design

DATA ANALYSIS

- Most common issue
 - Bus appears stopped for no reason
- Most requested feature
 - Bus status
- Most useful notification
 - Bus approaching stop
- Most people that do not use a transit app do not because they are frustrated
 - This issue leads to lost trust
 - Most say they would use a transit app if it were more transparent

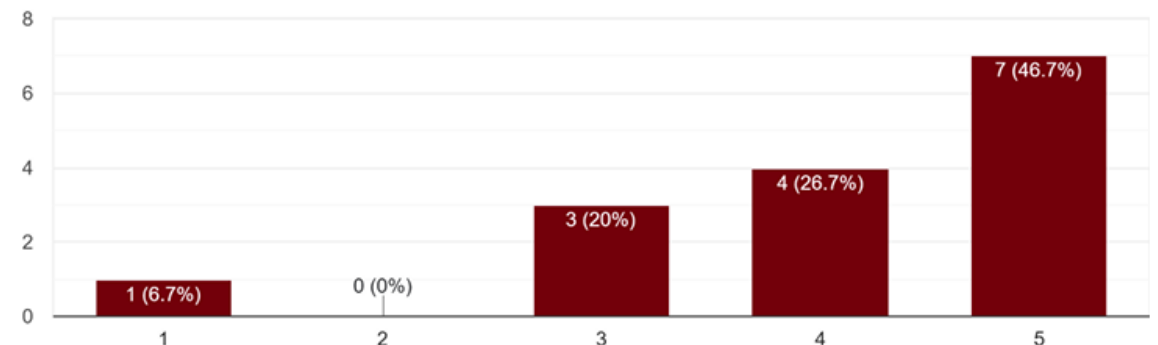
7. How do you currently feel when a bus appears stopped with no explanation?

15 responses



9. How likely would you be to use a transit app that clearly explains bus status and service interruptions?

15 responses



USER STORIES

- Based on real user needs from our research
- Focused on helping users make quick decisions
- Includes both riders and drivers
- Examples:
 - Students need clear bus status (in service, delayed, on break) to decide whether to wait or walk
 - Frequent riders want to favorite routes and receive alerts instead of checking constantly
 - Users want to enter start and destination and quickly compare bus vs walking
 - First-time riders need simple, easy-to-understand information
 - Drivers need to update bus status so riders get accurate information

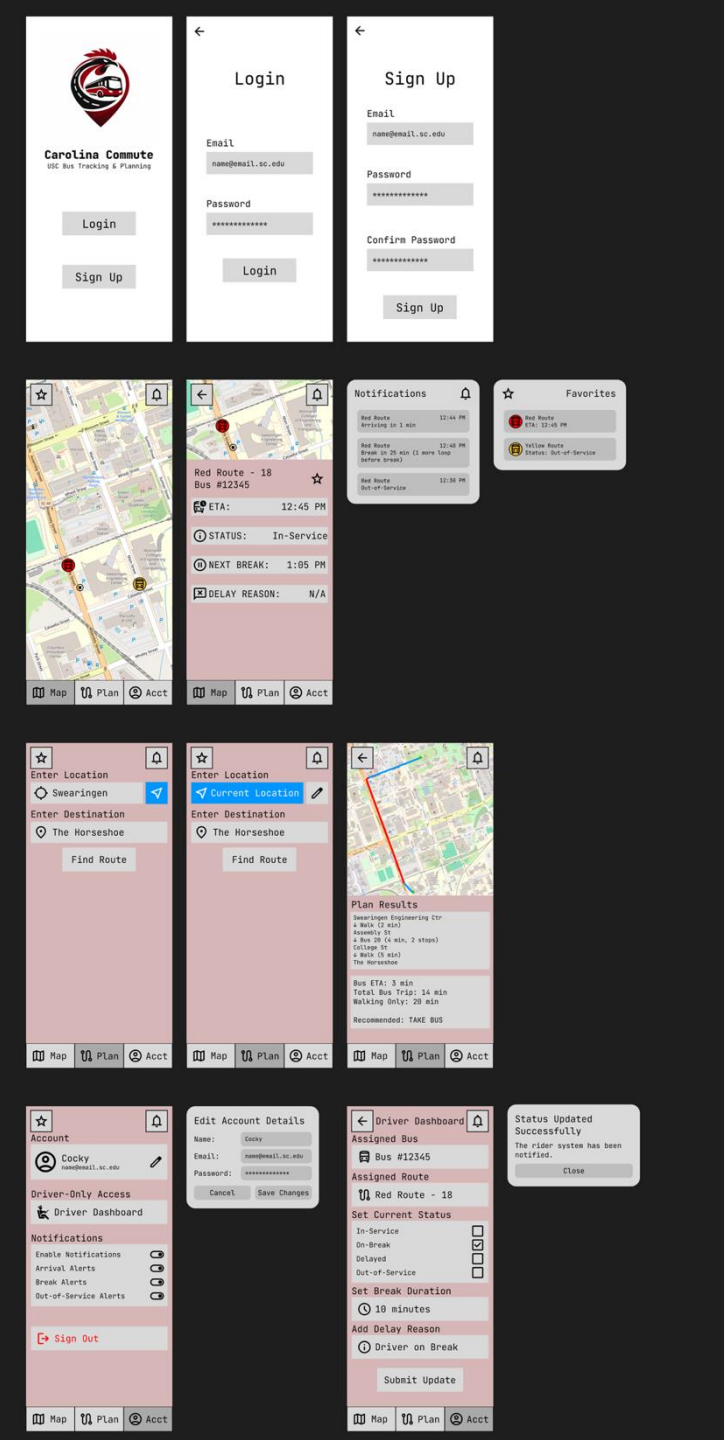


USER MAPPING

Activities:	Create / Manage Account		Plan Trip (Point A to Point B)		View Bus Status (Home)		View Alerts (Inbox)	Update Bus Status (Driver-only)	
Steps:	Create Account	Login + Account	Enter Trip Info	Compare Bus vs Walk	Display Map / List	Select Bus	View Inbox	Driver Login	Update Status
Details:	Enter email & password	Enter email & password	Input current location & destination	Retrieve bus ETA & walking time	Retrieve route & bus data	Display bus status & ETA (in service, break, delayed, out of service)	Show arrival, break & service alerts	Verify privilege & assigned bus	Select status type (in service, on break, delayed, out of service)
	Save credentials & create account	Verify credentials & log in	Identify closest stops (near origin & destination)	Highlight fastest option	Plot routes & active buses	Display delay reason or break time (traffic, driver change, detour)	Show alert timestamp	Navigate to Driver Dashboard	Enter break duration or delay reason
		Edit profile & sign out	Check route availability	Display combined walking + bus route	Toggle map/list & filter favorites	Indicate tracking unavailable (last known location)			Submit & confirm update
		Manage notifications & favorites (global toggle, arrival, break, out-of-service alerts, manage/remove favorites)	• Breaks the app into Activities → Steps → Details • Shows how the full app flow works • Connects what the user does to what the system does • Example: • Planning a trip → enter location → compare options / view route						Push update to rider system

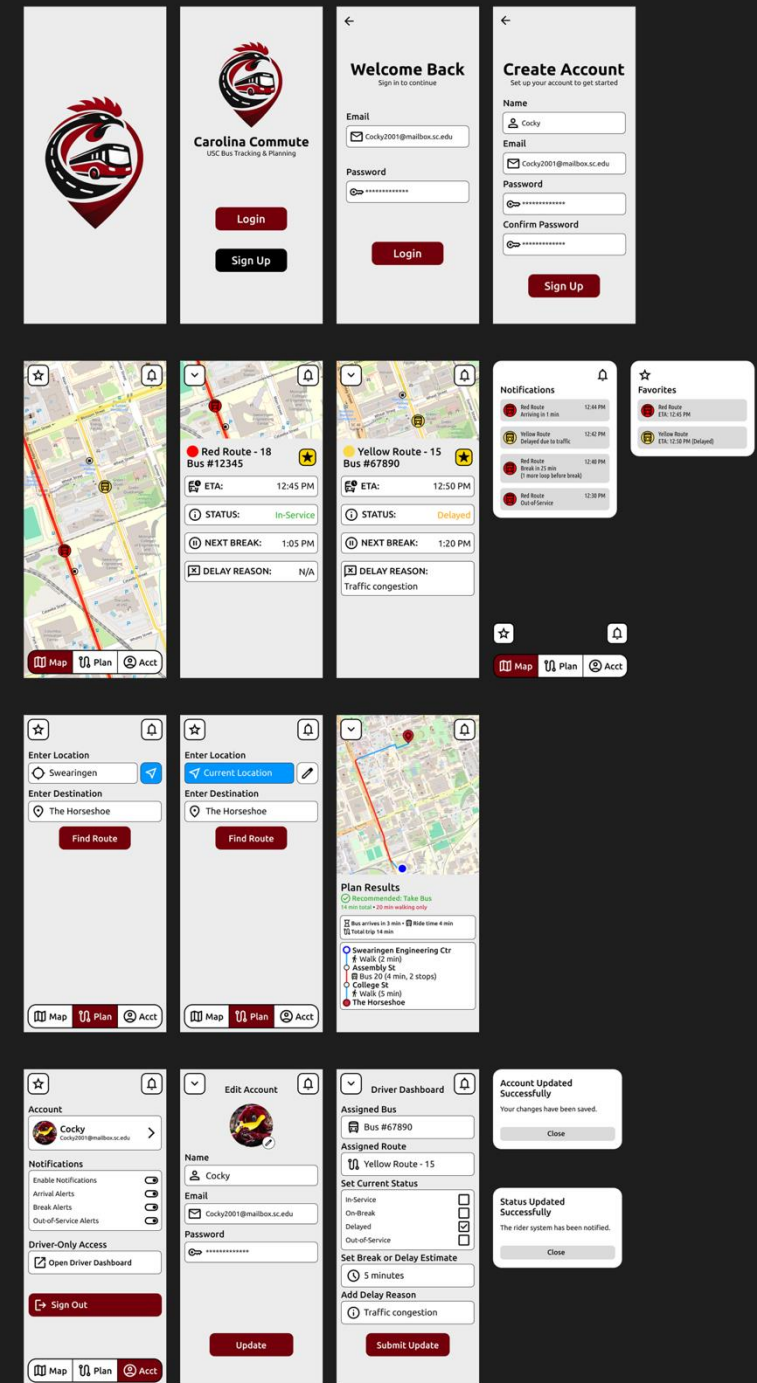
LOW-FIDELITY PROTOTYPE

- Focused on layout and functionality, not visual design
- Used simple styling to test structure and flow
- Included core features early:
 - Map with bus tracking
 - Route planning
 - Account + driver dashboard
- Purpose:
 - Test navigation and user flow
 - Identify missing steps or confusion



HIGH-FIDELITY PROTOTYPE

- Refined and polished the interface
- Improved consistency across all screens
- Replaced placeholder styling with final design system:
 - Clean gray background
 - Consistent colors and typography
- Improved readability and hierarchy:
 - Better spacing and alignment
 - Clearer grouping of information
- Adjusted certain interactions:
 - Changed edit account from overlay → full screen for better usability



USABILITY TESTING



Methods:

Survey (19 participants)
In-person observations /
interviews (both in and out
of class)



Tasks:

Check bus status
Plan route
Bus driver update



Results:

High success rates
Avg ratings ~4.3–4.6 / 5



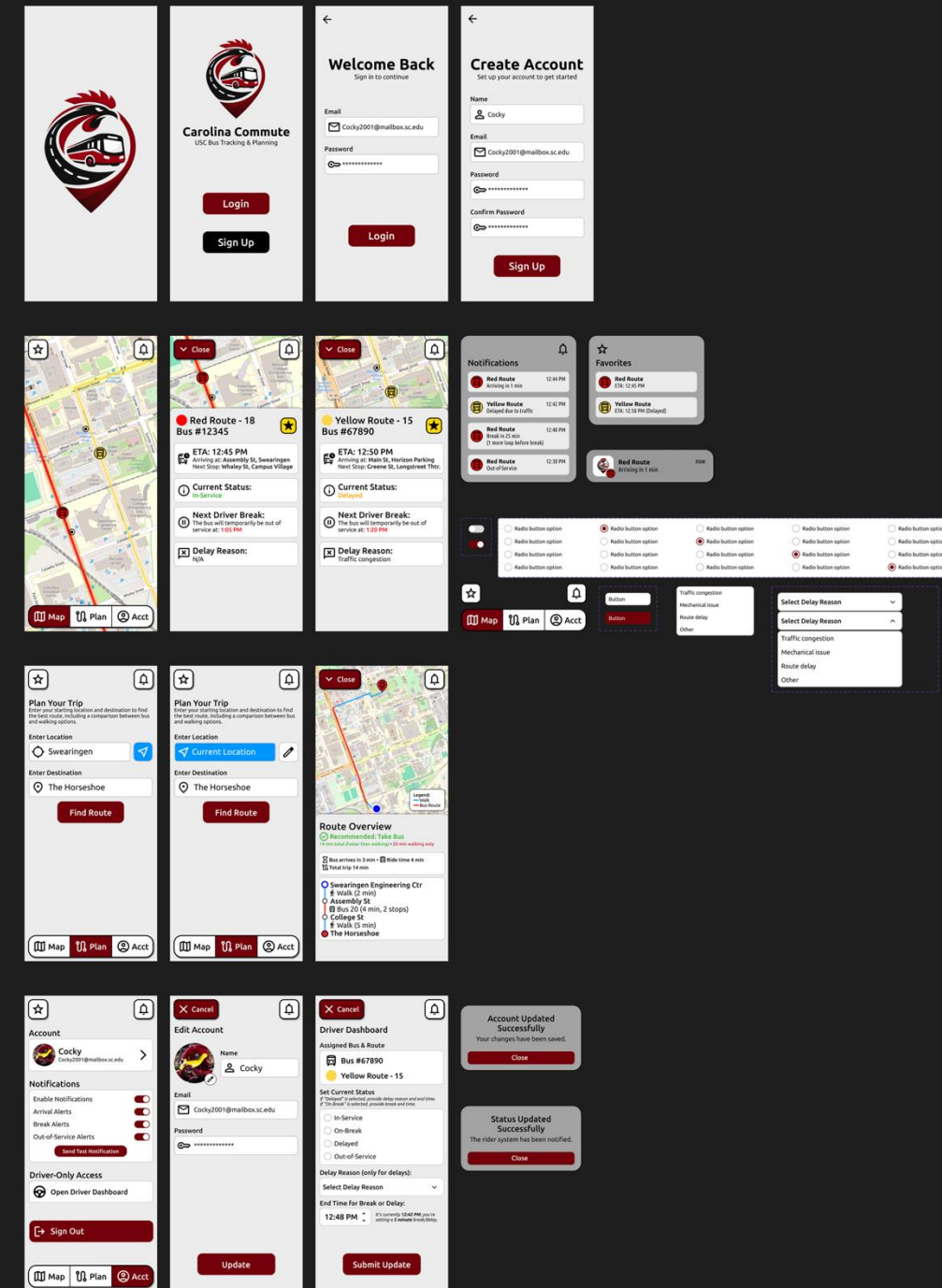
Key feedback:

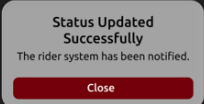
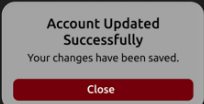
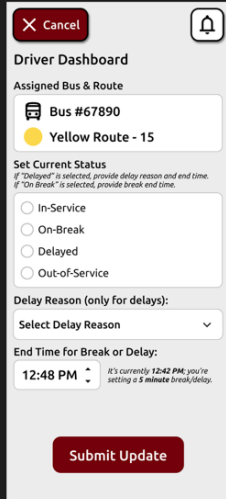
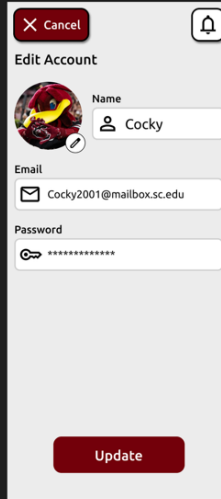
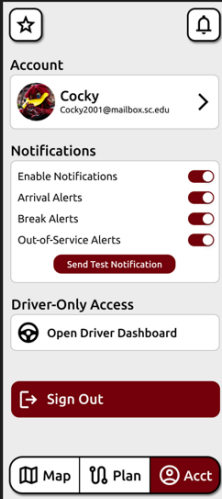
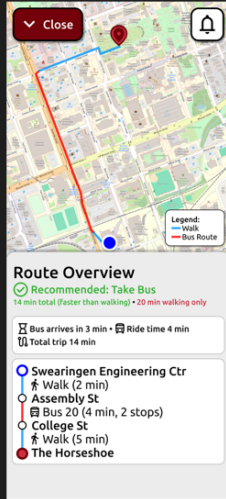
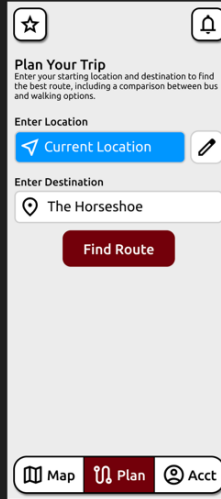
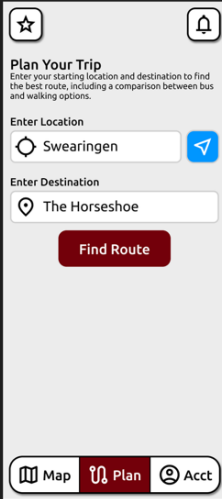
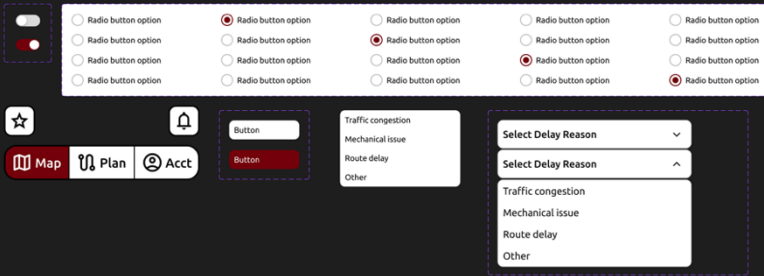
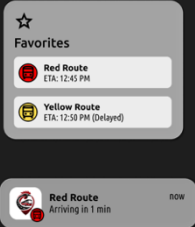
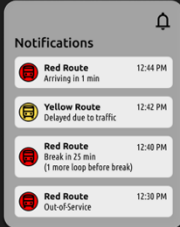
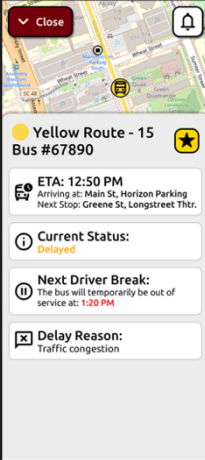
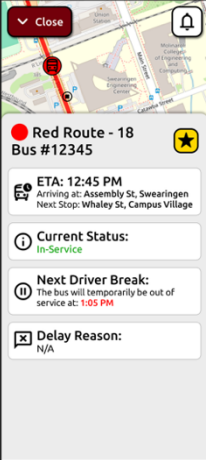
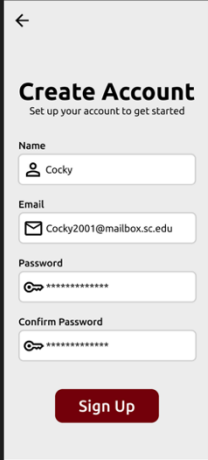
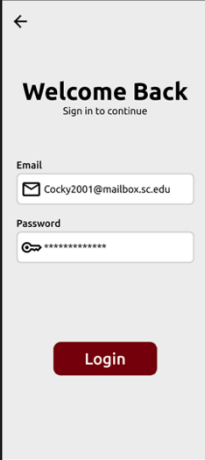
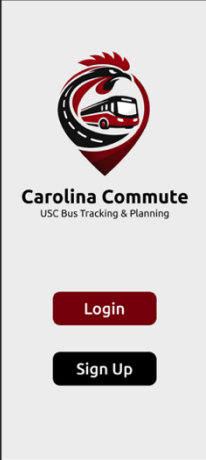
Unclear what ETA referred to
Need for bus stop context
Confusion with route colors
Minor issues with driver
dashboard inputs
Navigation could be clearer
in some areas

HIGH-FIDELITY PROTOTYPE UPDATE

Figma Link

- Changes based on feedback:
 - Added next stop to clarify ETA
 - Added route legend (walk vs bus)
 - Improved driver dashboard inputs
 - Improved navigation
 - Added interactive notification toggles
 - Added simulated notification behavior
- Focus:
 - Improve clarity without overcomplicating the design
 - Make small changes that have a big impact





EMERGING TECHNOLOGIES



Transit data API (GTFS — The COMET)

Standard format for routes, schedules, and real-time bus updates



GPS / device location

Allows users to use current location for route planning



Push notifications

Alerts for delays, breaks, and arrivals



Impact:

Enables real-time tracking
Helps users make faster decisions



Considerations:

Data accuracy depends on API
Reliability of real-time updates

DEVELOPMENT TEAM HANDOFF

Implementation priorities:

- Build bus status and route planning first
- Focus on core features before adding extras

Key features:

- Clear ETA with next stop
- Real-time bus updates
- Notifications for delays and alerts

Development notes:

- Clearly define how features and interactions behave
- Include edge cases (delays, errors, no data)

Risks:

- Accuracy of external API data
- Delays in real-time updates

THANK YOU!